Preliminary Draft Technical Note Document for Technical Specification Discussion for Coil Yard & Slab Yard Management System

for

Hot Strip Mill, JSOL, Angul

(Ref No - 26/JSOL/CYMS-SYMS/v0)

07th Jan 2026 Ver. 0

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This document is only for discussion purposes and not for contracting or placing order (PO).

Revision History

Sr. No.	*Revised By*	*Checked By*	*Approved By (QMS)*	*Details*	*Date*	*Rev No.*
1	*V. Tiwari*	*S. Randive*	*M. Patil*	• *First issue*	*08-01-2026*	*0*

ABBREVIATIONS

HIL: Hitachi India Pvt. Ltd.
SV: Supervisor
HMI: Human Machine Interface
PLC: Programmable Logic Controller
L1: Level-1 system
L2: Level-2 system
CYMS : Coil Yard Management System
SYMS : Slab Yard Management System

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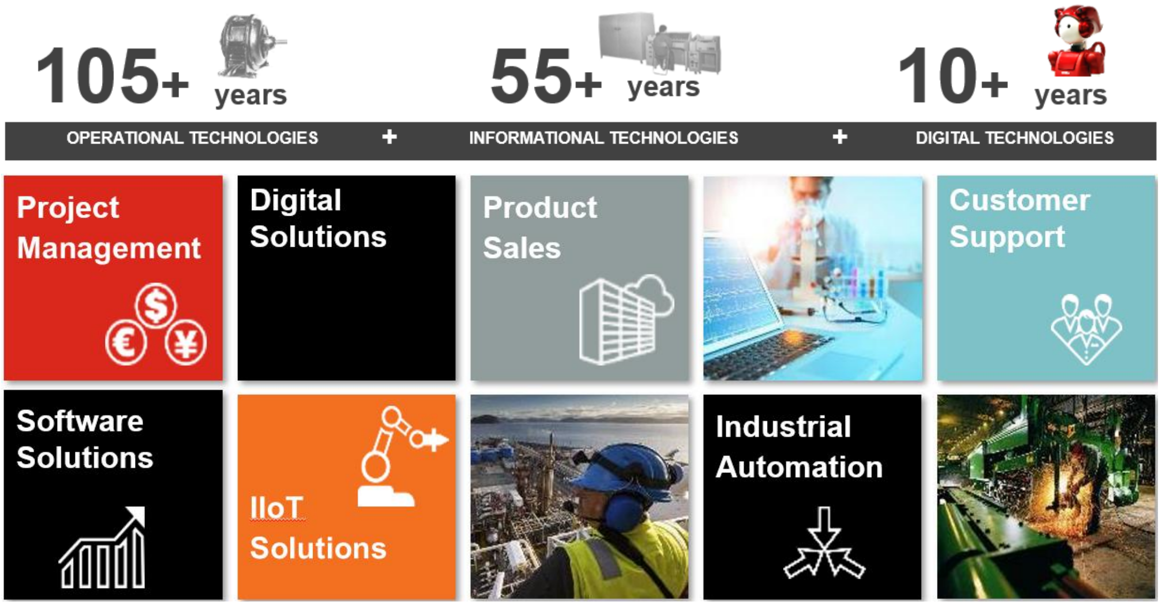
1. EXECUTIVE SUMMARY

Hitachi India Pvt. Ltd. (hereafter also referred to as HIL) provides automation solutions and digital solutions designed and developed for the industrial sector, particularly the metals industry. It takes advantage of more than 100 years of OT (Operation Technology) experience with process knowledge experts & SMEs and 55+ years of IT experience with the newest automation technology.



HIL is uniquely qualified to provide the control system and digital solutions you desire due to our in-house team of well-trained and certified professionals with the right experience & knowledge in designing, developing, and deploying the best solutions possible over a wide range of sectors & areas.

Some of our clients include:

TATA	JSW Steel	सेल SAIL
ADITYA BIRLA HINDALCO	AM/NS INDIA	posco

2. GENERAL OVERVIEW

This document is a Draft Technical Note for Design, Engineering, Supply and Commissioning of Coil Yard & Slab Yard Management System (CYMS & SYMS) at HSM, JSOL., Angul. This Technical Note is in response to the email enquiry request from JSL, dated 17th December,2025 and subsequent discussions.

Hitachi India Pvt. Ltd. (hereafter SELLER) is pleased to submit this Coil Yard & Slab Yard Management System (CYMS) & (SYMS)draft technical note (for discussion) for Hot Strip Mill (HSM) at JSOL, Angul (hereafter BUYER).

HIL is offering a solution that we are confident meets Buyer's Coil Yard management requirements and will bring significant value for Plant Managers.

It also sets the stage for a phased Digitalization, IT & business process maturity program, AI based projects, Centralised Plant Dashboard which aims at bringing alignment across the business & manufacturing processes.

Advantages of data visualization can be viewed in terms of direct business benefit or improvement, operational efficiency or productivity, and IT efficiency.

3. Offerings

3.1 Overview of the Coil Yard & Slab Yard Management System (CYMS & SYMS)

This technical note outlines the draft technical requirements for implementing a Coil Yard & Slab Yard Management System (CYMS & SYMS) in a steel plant, where overhead cranes operate in manual as well as Auto mode. The CYMS & SYMS will provide real-time coil location tracking, storage management, and coil/Slab retrieval support, using position sensors installed on the cranes by Buyer.

System Objective:-

- Ensure real-time, accurate tracking of Coil & Slab locations in the yard
- Avoid duplication, misplacement, or manual errors
- Provide operator and supervisor visibility of coil & slab inventory
- Enable efficient coil & slab storage and retrieval operations
- Log crane movement and coil & slab pick/drop events

System Architecture:-

a. Crane Subsystem (Buyer Scope and to be newly added as per requirements)

- Manually as well as Automatically operated cranes
- Position sensors (X, Y and Z axis)
- Pick/drop detection sensors
- Onboard controller
- HMI for operator feedback

b. CYMS & SYMS Server & Software:
- Digital yard map
- Real-time coil tracking
- Manual & Auto override and history logging
- API support

c. Yard Layout Configuration:
- Grid layout (zones, rows, columns)
- Configurable in software

Crane Hardware Requirements (Crane Hardware is in Buyer Scope)

• Position Sensors: ±100 mm accuracy
• Coil Pick/Drop Detection: Load/magnet-based
• Crane Controller: Edge I/O + network
• CYMS & SYMS Server: Industrial PC
• Network: Wi-Fi or wired (to be discussed during detailed engineering)
• HMI

CYMS & SYMS Features

- Live Tracking
- Intelligent Coil and Slab Storage and Retrieval
- Event Logging
- Ease of Traceability
- Yard Visualization
- Search & Retrieval
- Manual Corrections
- Reports
- Slab Yard and Coil Yard Modification and Configuration option
- 3D visualization
- Instruction Generation & Management
- Screen & Dashboard Modification option
- Smart Inventory Management

System Logic

a. Coil & Slab Placement Logic:
- Manual & Auto placement
- Position sensors detect X, Y, Z
- Coil/Slab Pick/Drop sensor logs event
- CYMS & SYMS records location

b. Coil/Slab Retrieval Logic:
- Select coil/Slab manually/Automatically
- Crane moves to location
- Pick sensor logs event
- CYMS/SYMS updates slot

Integration

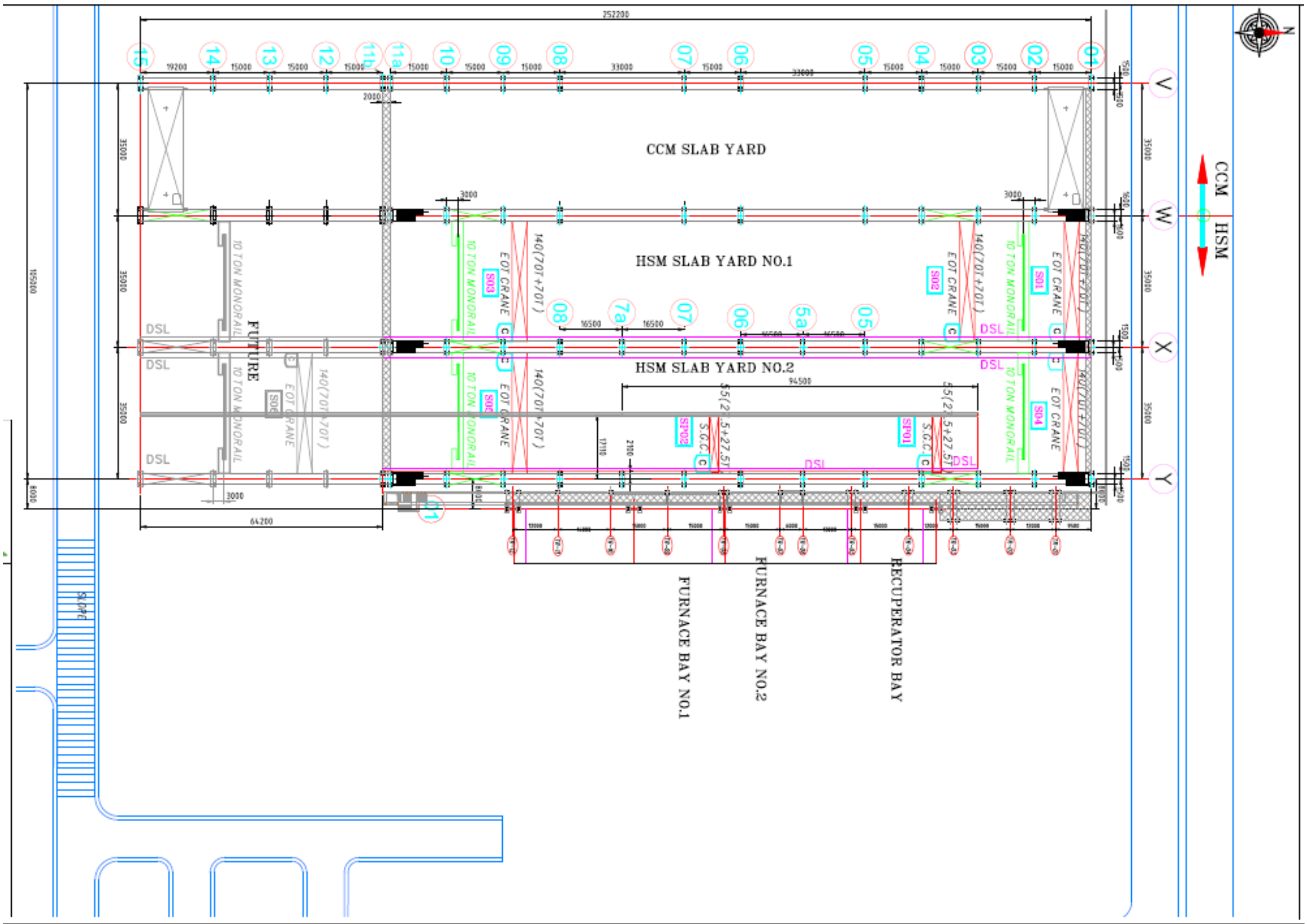
- HSM L1:- Coil Tracking Map
- MES/ERP: Coil data(Future Scope)
- Cranes (EOT and Semi Gantry)
- Mill Level 2.
- Mill Level 3.
- Caster 1 L1
- Caster 1 L2
- Slab Transfer Car
- Caster 2 L1
- Caster 2 L2
- Pellet Conveyor L1
- Pellet Conveyor L2

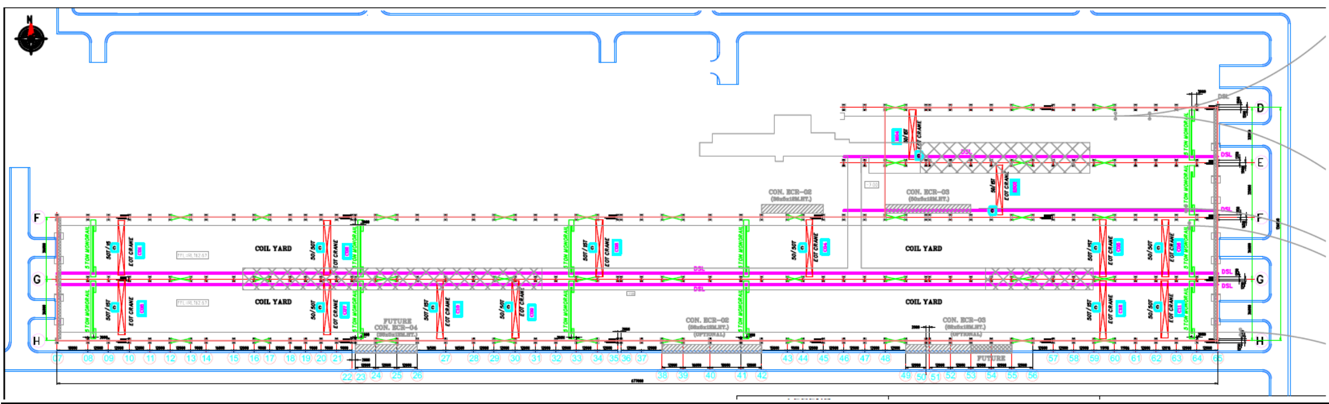
Yard and Cranes to be considered:-

Slab Yard Map details as provided by Buyer are as under.

Calculation		WX Bay	XY Bay
No. of piles possible in a row.		2	2
No.of rows possible in a bay.		47	47
Total No. of piles in a bay.		94	94
Total No. of slabs in a bay.		1410	1410
Total Tonnage capacity of Slab Yard	78960		
Total no. of Slabs stored	**2820**		

Calculation		FG Bay	GH bay	DE bay	EF bay (Column 48 to 65)
No.of rows possible in Coil yard		9	9	8	8
No.of Coils possible in a row		212.38	212.38	53.97	53.97
Total No. of Coils in a Bay		1911.43	1911.43	431.75	431.75
Total tonnage of storage in a bay		53520	53520	12089	12088.89
Total Tonnage capacity of coilyard	131217.78				
Total no. of coils	**4686.35**				
Considering 2 HI					
Bottom Row	**5**				
Top Row	**4**				
No.of Coils possible in a row		382	382	97	97
Total No. of Coils in a Bay		3438	3438	776	776
Total tonnage of storage in a bay		96264	96264	21728	21728
Total Tonnage capacity of coilyard	**235984**				
Total no. of coils	**8428**				





Crane Information:-

Existing crane details as provided by the Buyer are as below.

Summary		
Description	**Quantity**	**Yard Location**
Slab Yard EOT Crane	5	Slab Yard
Slab Yard Semi Gantry Crane	2	
Slab Yard transfer Car	1	
Coil Yard EOT Crane	15	Coil Yard
Forklift	6	Coil Yard

Details					
CRANES & EQUIPMENT LIST FOR HSM PROJECT					
S.No.	**DESCRIPTION AS PER PRIORITY**	**Rev Qty**	**Crane Name**	**BAY**	**Yard Location**
1	Slab Yard 2 - 140 MT	1	Crane no S04	XY BAY	Slab Yard
2	Slab Yard 1 140 MT	1	Crane no S01	WX BAY	Slab Yard
3	Slab Yard 1 140 MT	1	Crane no S03	WX BAY	Slab Yard
4	Slab Yard 2 140 MT	1	Crane no S05	XY BAY	Slab Yard
5	Slab Yard 1 - 140 MT	1	Crane no S02	WX BAY	Slab Yard
6	Slab transfer Car	1	Transfer Car 1	VW-WX-XY BAY	Slab Yard
7	Slab Pick Up Bay Semi Gantry 55MT	1	Crane no SP01	XY BAY	Slab Yard
8	Slab Pick Up Bay Semi Gantry 55MT	1	Crane no SP02	XY BAY	Slab Yard
9	Coil Yard 50/15 MT	1	Crane No C1	FG BAY	Coil Yard
10	Coil Yard 50/50 MT	1	Crane No C2	FG BAY	Coil Yard
11	Coil Yard 50/15 MT	1	Crane No C3	FG BAY	Coil Yard
12	Coil Yard 50/50 MT	1	Crane No C4	FG BAY	Coil Yard
13	Coil Yard 50/15 MT	1	Crane No C5	FG BAY	Coil Yard
14	Coil Yard 50/50 MT	1	Crane No C6	FG BAY	Coil Yard
15	Coil yard 50/50 MT	1	Crane no C7	GH BAY	Coil Yard
16	Coil yard 50/15 MT	1	Crane no C8	GH BAY	Coil Yard
18	Coil yard 50/15 MT	1	Crane no C9	GH BAY	Coil Yard
19	Coil yard 50/50 MT	1	Crane no C10	GH BAY	Coil Yard
21	Coil yard 50/50 MT	1	Crane no C11	GH BAY	Coil Yard
22	Coil yard 50/15 MT	1	Crane no C12	GH BAY	Coil Yard
23	Coil yard 50/50 MT	1	Crane no RS03	EF BAY (roll shop Bay)	Coil Yard
24	Coil Yard 50/15 MT	1	Crane no M04	DE BAY	Coil Yard
25	Coil Yard 100/45 MT	1	Crane no M03	DE BAY	Coil Yard
26	Coil Yard Forklift	6	Fl01-FL06	DE--EF-FG-GH BAY	Coil Yard
	Total	**29**			

Slab data	**Value**	**Unit**
Slab weight, max.	36	Ton
Slab length, max.	12000	mm
Slab length, min.	4500	mm
Slab thickness, min.	180	mm
Slab thickness, max.	260	mm
Slab Width, min.	800	mm
Slab Width, max.	1680	mm
No.of Slabs in a pile.	15	No.s
Centre to Centre distance between 2 slabs	3500	mm
Minimum gap between 2 piles	2000	mm
Average weight of a slab	28	Ton

Coil data	**Value**	**Unit**
Coil weight, max	36	Ton
Coil diameter outer	1000 – 2150	mm
Coil width	800 – 1680	mm
Strip thickness	1.0 – 20	mm
Coil temperature max.	850	° C
Average Coil Dia	1600	mm
Average Coil Width	1300	mm
Max Coil Dia	2150	mm
Max Coil Width	1680	mm
Gap between 2 coils	1000	mm
Gap between 2 rows	1900	mm
Avg Coil Weight (T)	28	Ton

Complete Coil Yard

Coil Yard	FG Bay (Column 7 to 65)	GH bay (Column 7 to 65)	DE bay (Column 48 to 65)	EF bay (Column 48 to 65)	
Length (in m)	677	677	178	178	
Width (in m)	33.5	33.1	30	30	
Effective Length (in m)	669	669	170	170	5 meters clearance for coil conveyor line.
Effective Width (in m)	31	31	30	30	2.5 metres clearance for railway track
Area (in sqm)	22679.5	22408.7	5340	5340	
Effective Area (in sqm)	20739	20739	5100	5100	
Cranes EOT	6	6	2	1	
Cranes Capacity (in Ton)	50/15 MT -3 & 50/50 MT-3	50/15 MT -3 & 50/50 MT-3	100/45 MT & 50/15 MT	50/15 MT	
Crane Names	C01,C02,C03,C04,C05,C06	C07,C08,C09,C10,C11,C12	M03,M04	RS03	
Semi Gantry Crane	NA	NA	NA	NA	
Capacity	NA	NA	NA	NA	
Crane Names	NA	NA	NA	NA	
Row along railway track	F Row (Column 7 to 65)	H Row (Column 7 to 65)	D Row (Column 48 to 65)	F Row (Column 54 to 65)	
Conveyor Length					
Length of Track near coil yard	677	677	178	139	
Fork Lift	2	2	2	2	

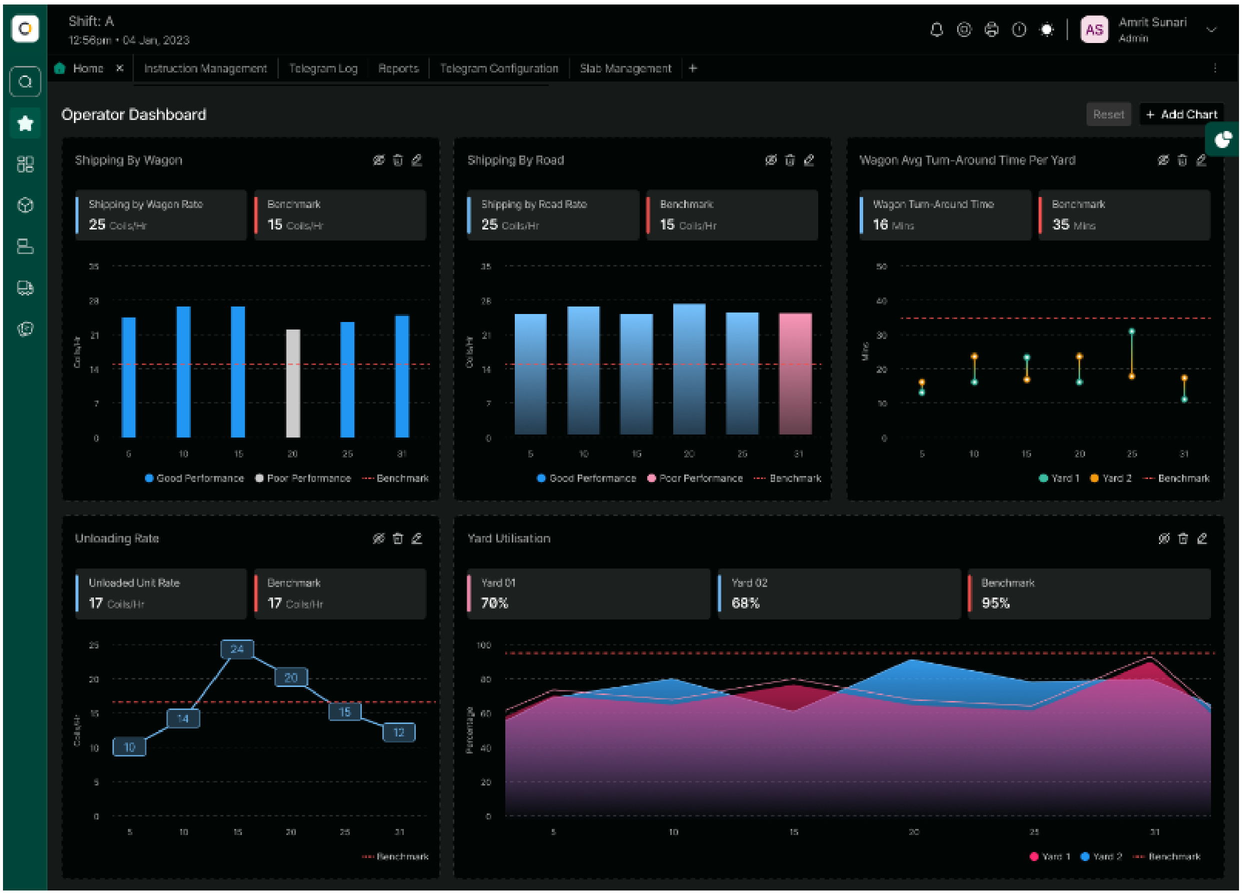
Value Added Items in the CYMS & SYMS:

1. This system includes software-based Anti-Collision alarms for the user.
2. Zone-based crane movement provision (handled on software side).
3. Crane performance KPIs Monitoring on Dashboard (like yard turnaround time, utilization, idle rate tracking).
4. Crane work order management for optimum crane movements/logistics for energy savings/time savings
5. Optimized algorithm for coil & slab stacking & storage in coil & slab yard (from safety aspects).

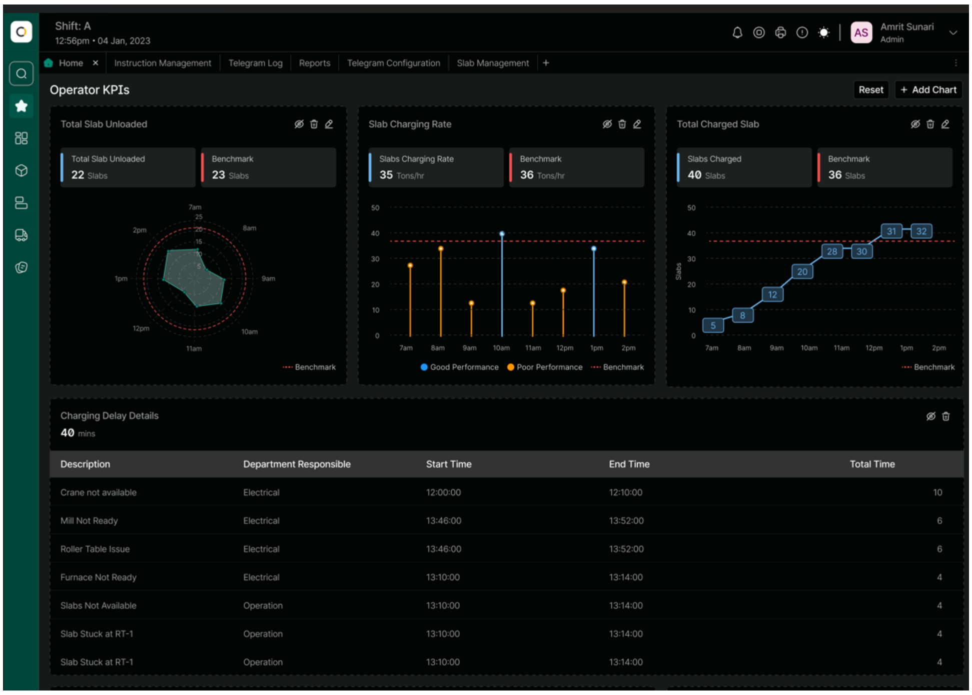
Reference Screens:-

Following are some screenshots of CYMS & SYMS just for reference purposes. Please note that CYMS & SYMS system is fully customized and tailormade as per the site requirements and Buyer’s needs decided mutually between Seller and Buyer.

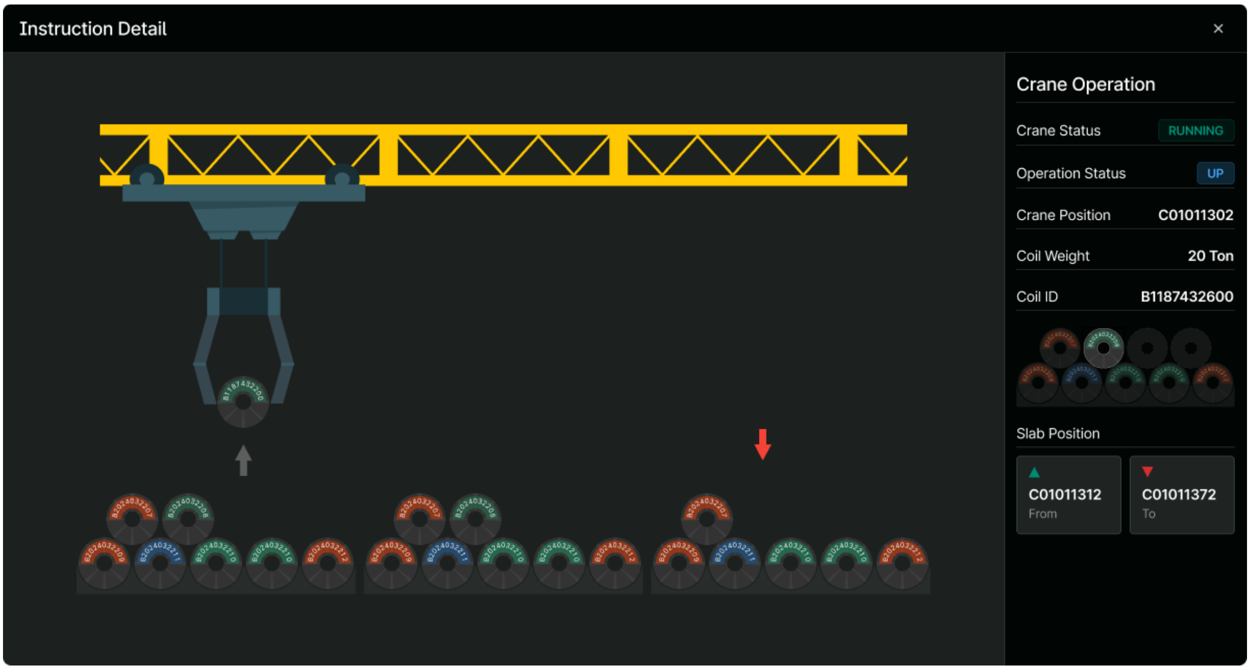
Operator Dashboard Screen:



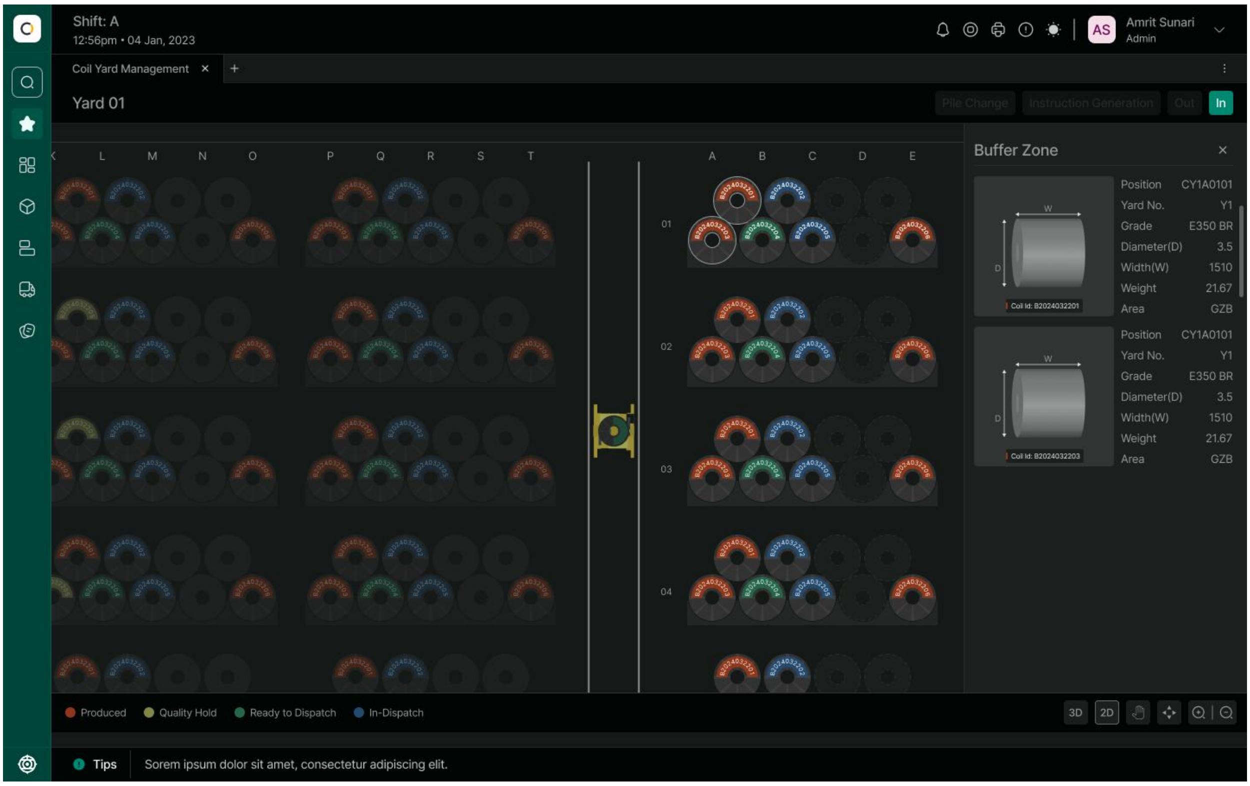
Operator Monitoring Dashboard:

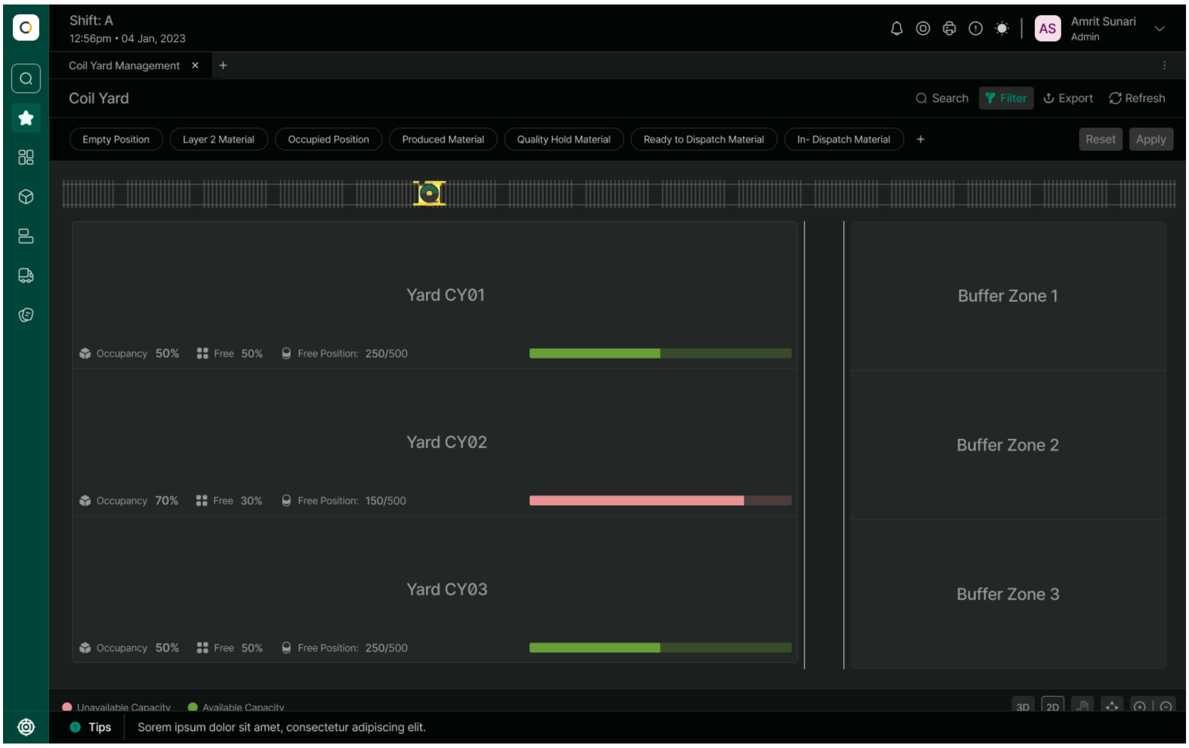


Instruction Detail Screen:



Yard Map Details Screen:





Remote Support System

Supplier will provide complimentary remote maintenance support for troubleshooting and issue resolution (if any) for a period of 6 months from the date of completion of project. Necessary infrastructure and network configuration will have to be enabled by buyer to facilitate this. Remote maintenance support will be limited to scope of supply of this Technical Note . Also, access to remote terminal should be made available to Seller as per requirement. Dedicated Internet/Lease line at site to be arranged by Buyer.

The effective working hours at HIL Office shall be from 9:00 AM to 5:00 PM IST (Monday to Friday), excluding National Holidays.

NON-DISCLOSURE AGREEMENT (NDA) will be signed between Buyer and Seller before starting the remote support.

3.1.1 Documentation Control

Supplier shall provide documentation with respect to Screen Design Document, Hardware Specifications, software specifications, CYMS & SYMS Application Operation Manual.

3.1.2 Customer Training

Suppliers shall provide training at site during commissioning to a maximum of 5 persons for a maximum of 3 days. Training shall cover mutually agreed topics on CYMS & SYMS Web application supply. Training shall comprise of classroom training at site.

4. TECHNOLOGY STACK

The technology stack for various components is as follows:

Components	Technology Used
Application Front-end	Angular Framework. Application can be viewed on a standard web browser like Chrome, Edge & Mozilla
Database	MS SQL Server 2022 Standard Edition
Application Back-end	Python Django
Docker	For Deployment

4.1 BASIC HARDWARE SPECIFICATIONS

Following is the list of Hardware required for CYMS & SYMS Application.

SR. NO.	EQUIPMENT NAME	SPECIFICATIONS	MAKER	SERVER QUANTITY (NOS.)
1	Online and Backup Server	• Intel Xeon-Gold 6526Y 2.8GHz 16-core 195W Processor for • 32GB (1x32GB) Dual Rank x8 DDR5-5600 CAS-46-45-45 EC8 Registered Smart Memory Kit • 960GB SATA 6G Mixed Use SFF BC Multi-Vendor SSD	HPE/ DELL/ LENOVO	4
2	Online and Backup Server Console & accessorie s	S5 Pro 524pf FHD Monitor with USB Mouse and Keyboard	HPE/ DELL/ LENOVO	4
3	KVM Extender Set	CE100: USB VGA CAT 5 Mini KVM Extender (1280x1024@100m) along with VGA Cable.	ATEN	4
4	CYMS and SYMS Client PCs(4 Each)	Windows 11, i7 Processor 16 GB RAM 500 GB Memory USB Keyboard and mouse	HP/Dell/L enovo	8

4.2 BASIC SOFTWARE SPECIFICATION

Below are the Software Specifications for the Proposed CYMS system.

SR. NO.	EQUIPMENT/SOFTWARE NAME	MAKER	QUANTITY (NOS.)
1	Windows Server 2022 (64 bit) (5 CAL)	Microsoft	4
2	Microsoft Office 2021(For Servers and Client)	Microsoft	12
3	Microsoft Visual Studio 2022	Microsoft	4
4	Backup software	Acronics	4
5	Antivirus Software	Trend Micro	12
6	Python	-	4
7	Django Framework	-	4
8	Angular Framework	-	4
9	MS SQL Server 2022 Standard Edition with 5 CAL	Microsoft	4
10	Docker		4

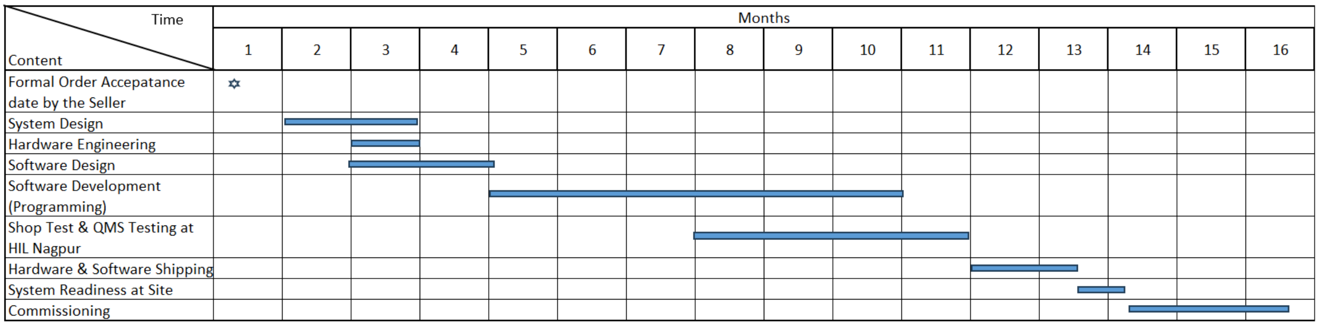
4.3 THIRD PARTY SOFTWARE REQUIREMENTS

Database: SUPPLIER CYMS system uses MS SQL Server 2022 Standard Edition OR other suitable database. Decision regarding selection of database is solely decided by SUPPLIER.

Remote Link: To provide a suitable level of response to operation & process execution problems and queries raised on site, Seller requires a network connection via broadband / VPN to be established between Seller and the Buyer’s site. Set-up and maintenance of this link will always be under the buyer’s authority.

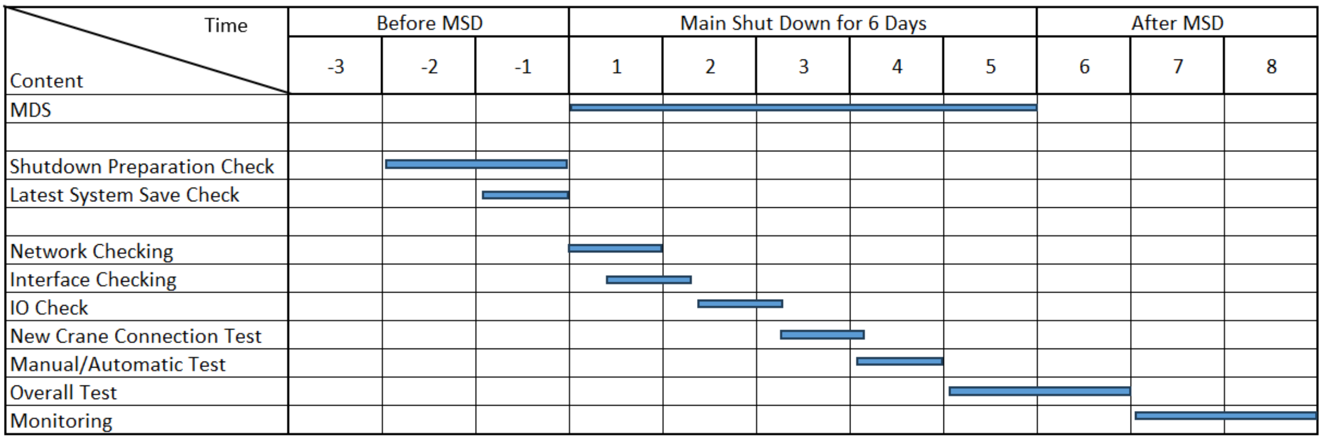
5. SCHEDULE

5.1 OVERALL SCHEDULE INDICATIVE FOR REFERENCE ONLY



Note: After Approval on System Design Document Seller will take 8 Months for Software development & Testing. In the event of a delay in System design document approvals from the Customer, it will lead to an overall delay in the delivery. Above delivery schedule is for CYMS & SYMS application

SHUTDOWN SCHEDULE INDICATIVE FOR REFERENCE ONLY



The above shutdown schedule provided is for reference only. The final shutdown schedule will be determined through discussion and mutual agreement between the BUYER & SELLER.

5.2 Supervisors:

The following site-supervisor will be deputed to the site for the commissioning, deployment & training at site:

- Project Manager: 20 Days
- CYMS Developer: 20 Days
- SYMS Developer: 20 Days
- Commissioning SV (QA SV): 60 Days

Total 120 working days (Inclusive of on-site training).

6. SCOPE OF SUPPLY

6.1 SCOPE OF SUPPLY DEFINITIONS

The engineering and software development service necessary for the proposed Coil Yard & Slab Yard Management System shall be provided either by Buyer or by the Seller in accordance with the Scope of supply List. The matters not described in the said list shall be determined through mutual discussion.

Buyer : JSOL (B: Buyer)

Seller : Hitachi India Pvt Ltd (S: Seller)

BD : Basic Data
BE : Basic Engineering
DD : Detail Design
SU : Supply or Service
ER : Erection
COM : Commissioning

X/Y: Primary Work/Services provided by X with Support from Y

6.2 DIVISION OF ENGINEERING AND SOFTWARE DEVELOPMENT ERECTION/ COMMISSIONING SERVICES OR WORKS

No.	ITEM	Responsibility					
		BD	BE	DD	SU	ER	COM
(1)	**Services**						
-1	Project Execution	B/S	B/S	B/S	-	-	-
-2	Overall system design	S	S/B	S/B	-	-	-
-3	Work Test and Simulation	S	S	S	-	-	-
(2)	**SYSTEM Engineering**						
-1	Documentation	S	S	S	S	-	-
(3)	**HARDWARE**						
-1	**Online Server**	S	S	S	S	B	S
-2	**Backup Server**	S	S	S	S	B	S
-3	**Application Servers Console and Accessories**	S	S	S	S	B	S
-4	**KVM Extenders**	S	S	S	S	B	S
-5	**Network Cables**	B	B	B	B	B	B
-6	**Server Rack**	S	S	S	S	B	S
-7	**Client PCs**	S	S	S	S	B	S
(4)	**SOFTWARE**						
-1	Windows Server 2022 (64 bit) (5 CAL)	S	S	S	S		S
-2	Microsoft Office 2021	S	S	S	S		S
-3	Microsoft Visual Studio 2022	S	S	S	S		S
-4	Backup software	S	S	S	S		S
-5	Python	S	S	S	S		S
-6	Django	S	S	S	S		S
-7	Angular Framework	S	S	S	S		S
-8	Docker	S	S	S	S		S
-9	MS SQL Server 2022 Standard Edition(5 CAL)	S	S	S	S		S
-10	Antivirus Software	S	S	S	S		S

(5)	**TRAINING**						
-1	Training (Max 3 days) for 5 persons on CYMS & SYMS	S	S	S	S	-	-
(6)	**Pre-Engineering Activities**						
-1	Study of Existing system	S	S	S	S	-	-
(7)	**Documents to be submitted for Reference & Records**						
-1	Screen Design Documents	S	S	S	S	-	-
-2	Hardware Specifications	S	S	S	S	-	-
-3	Software Specifications	S	S	S	S	-	-
-4	CYMS & SYMS Operation Manual	S	S	S	S	-	-
(8)	**Crane System**						
-1	Position Sensors (X and Y Position)	B	B	B	B	B	B
-2	Coil Pick/Drop Sensor	B	B	B	B	B	B
-3	Wireless Access Points	B	B	B	B	B	B
-4	Networking Accessories	B	B	B	B	B	B
-5	Crane Interface and Integration with CYMS (Crane side sensors/devices/PLC modifications)	B	B	B	B	B	B
-6	Crane Interface and Integration with CYMS & SYMS (CYMS & SYMS Software side modification)	S	S	S	S	S	S
(9)	**Interface with 3rd Party Systems**						
-1	3rd Party(As per section 3.1 Interface) Interface and Integration with CYMS & SYSM (for 3rd party side modifications)	B	B	B	B	B	B
-2	3rd Party(As per section 3.1 Interface) Interface and Integration with CYMS & SYSM (CYMS & SYMS Software side modifications)	S	S	S	S	S	S

Note: 1) Any additional requirements beyond the scope mentioned above shall be discussed and mutually agreed upon. A separate proposal will be submitted for such additional requirements.

2) Firewall Configuration will be managed by Buyer.

6.3 Work Completion Certificate

Upon the completion of the work (based on the below criteria), the Seller shall issue the **Work Completion Certificate** to the Buyer.

Work Completion Criteria:

- Supply of Hardware & Software as per the scope of supply (described in section 6.2)
- Deployment of CYMS & SYMS
- Commissioning work Man-days used (as described in section 5.2)
- Submission of all documentation as per the scope (described in section 3.1.1)

Upon completion of the above criteria, the Work Completion Certificate will be issued by the Buyer to the Seller, signifying the satisfactory completion of the project. The Buyer shall issue the Work Completion Certificate within seven (7) days of the completion of the above criteria. In case if the Work Completion Certificate is not issued by BUYER within this period, it is considered that Work Completion Certificate as deemed issued by the BUYER. If there are any issues/problems faced by BUYER, it will be mutually discussed, and SELLER would support BUYER with the best of the ability to resolve the issue.

By issuing the Work Completion Certificate, the Buyer acknowledges that the Seller has fulfilled all contractual obligations and requirements.

6.4 BUYER OBLIGATIONS

The Buyer should fulfil the following obligations

- Responsible for the project execution (answer technical queries in reasonable time and coordinate with all the stake holders of the project)
- Arrange all the hardware in Buyer scope well ahead of the agreed time schedule.
- Network cables & accessories not included in this Technical Note to be provided by Buyer
- Site access to Seller's representative for data collection and discussion with the technical team as per requirement.
- Dedicated internet connection
- In case of any health issue to Seller's representative, buyer to immediately provide best available medical facility. The expenses will be borne by the Seller.

EXCLUSION LIST

Only the services mentioned in the Scope of this are included. Any further services need to be mutually discussed and commercial / technical implications discussed and compensated through a previously agreed change management mechanism.

Specific attention is drawn to the following services that are excluded from the scope of this Technical Note(unless specifically included elsewhere in this).

In this Technical Note following are not included:

- Interface with external devices other than explicitly described in the Technical Note document.
- Software patches including but not limited to Microsoft Security updates, Antivirus updates or any other software upgrades or patches.
- Any warranty, guarantee, liability, responsibility, etc. about productivity, quality, yield, etc.
- Source code of software of core technology.
- Modifications to existing electrical and automation systems at HSM including but not limited to Level-1, Level-2, MES, SYC & FCE L1/L2 system.
- Erection activities wherever required for project execution. Any kind of Civil work.
- Hardware and software other than that is mentioned in technical document.
- Mechanical equipment supply, modification etc.
- Support for troubleshooting for mechanical/operation/process issues of customers.
- L2 & PDA software and related hardware
- Firewall and other networking components
- Performance with respect to productivity, yield, quality, process capability, process performance, etc.

6.5 Buyer Prerequisites:

- Readiness of the Crane System along with required hardware and Sensors
- Remote connection Set Up.

6.6 Binding Conditions:

- SELLER scope of supply & services as described in agreed & signed technical specifications (TS) shall be limited, as per the inputs/information obtained from the BUYER, only to the modification/creation of interface program for helping the BUYER to use SELLER supplied equipment/items for obtaining the required functioning at BUYER own responsibility.
- All the services, except for those under the scope of SELLER as per the agreed & signed TS, will be under the full responsibility of BUYER. SELLER neither guarantees any performance nor SELLER is liable for any claims for the non-performance of such Equipment & services.
- SELLER scope of supply & services excludes any supervision, management, regulation, arbitration and/or measurement of BUYER's personnel, agents or contractors and work related thereto nor does include any responsibility for the scheduling, monitoring or management of their work.

- SELLER supervisors shall have the right to leave the site without any permission or approval of BUYER in the following cases:
  a. If the idling periods extend more than a considerable time as judged by SELLER.
  b. If the BUYER does not follow the conditions of the agreed terms as per the agreed and signed TS
  c. An act of God such as natural calamities, war, riot, rebellion, invasion, commotion, epidemics, terrorism, or such other phenomenon which is predictable to affect the body or life of our supervisors directly or indirectly.
  d. If BUYER prerequisites are not fulfilled as per the agreed and signed TS.
  e. Circumstances at project site are not conducive and are beyond SELLER's control as judged by SELLER.
- SELLER will not be responsible for
  a. Performance guarantee of any kind.
  b. Productivity AND/OR Quality of the Product
- SELLER will not be responsible for any failed/missing/damaged equipment, item, and components etc. during the execution of the service work unless otherwise explicitly mentioned in the contract.
- Unless explicitly mentioned in the agreed and signed TS, SELLER will not take responsibility for any modification or updating of the existing system.
- Ensuring Normal working of existing servers/systems, L2 System, Crane System, Crane control, Crane instrumentations/sensors/devices and L1 system lies with Buyer.
- Responsibility of Firewall configuration and its proper working for Cyber Security aspects lies with Buyer.
- The effective date of execution of the contract shall be the PO acceptance date by the SELLER.
- Completion of scope of services of SELLER as per agreed and signed TS by virtue of receipt of Work Completion Certificate/Deemed Issued Work Completion Certificate by SELLER will be prerequisite condition to take up any new/additional scope of work for HSM, JSL.
- Responsibility and Liability for Seller- Nil, except for the scope defined in this draft technical note document.

6.7 Cybersecurity Disclaimer

SELLER strongly recommends that security updates are implemented by Buyer as soon as they are available and that the latest versions are used. Use of versions that are no longer supported, and failure to apply the latest updates may increase the exposure to cyber threats. This application contains confidential information. It is the sole responsibility of end-user to maintain the confidentiality and proper usage of the information contained herein. The BUYER is responsible for preventing unauthorized access to the existing plant systems. The BUYER will be responsible for Cybersecurity, Information Security, Hardware Security, Personal Data Security, Overall Data security, Network security, equipment security. In no event shall SELLER be liable for any direct, indirect, incidental, consequential, or special damages arising out of or in connection with any cybersecurity incidents, data breaches, unauthorized access, or any other security-related events, regardless of the cause of action.

7. DISCLAIMER

7.1 SOFTWARE LICENSES

The use of the delivered software, or parts of the same, is strictly limited to the services provided. The buyer has the right to use and modify the software for normal use and for the lifetime of the system supplied by Seller. Any duplication (except for normal back-up purpose), distribution to third parties, sell or use in other plants is prohibited. It's also prohibited to make software accessible to third parties. All rights of the software remain with Seller.

'BUYER is responsible for maintaining and ensuring valid license for use of third-party software and legal use of third-party software used in this system. In no event shall Seller be liable for any issue or claim or damage arising out of use of any third-party software.'

7.2 CHANGES DUE TO TECHNICAL IMPROVEMENTS

Seller follows a policy of continuous improvement of design, manufacturing and software development and therefore reserve the right to supply systems differing in detail from what is described in this document.

7.3 CONFIDENTIALITY OF INFORMATION (Non-Disclosure Agreement)

According to the law, Seller considered this document to be a company secret and therefore prohibits any person to reproduce or disclose it in whole or part to third parties, without specific written authorization of Seller Management.

7.4 LIMITATION OF LIABILITY & CONSEQUENTIAL DAMAGES

With respect to any and all liability of HIL out of or in connection with this Technical Note notwithstanding anything contained herein to the contrary in no event shall HIL's total liability for any and all claims, damages or liabilities arising out of related to its performance of this Technical Note , including but not limited to any liquidated damages, costs for repairs, replacement, rectifications of defects, dismantling, etc., exceed one hundred percent (100%) of the HIL's Contract Price. In no event shall HIL be liable for any indirect or consequential damages, including but not limited to any loss of profit, loss of use, loss of Contracts, loss of production or business interruption. According to law, SELLER considers this document to be a company secret and therefore prohibits any person to reproduce or disclose it in whole or part to third parties, without specific written authorization of SELLER Management

8. FREE PROOF-OF-CONCEPTS

Seller is pleased to provide a **complimentary Proof of Concepts (PoCs)** along with the aforementioned system to showcase SELLER's latest offerings in Composite Automation & Digital Solutions(CADS) & Digital Transformation (DX) Solutions. The PoC will be conducted on the same server.

The solutions under the PoC will be deployed for a limited period of 2-months to demonstrate the capabilities of these solutions. Upon successful completion of the demonstration, the BUYER must capitalize on the benefits by purchasing these solutions through a separate commercial contract, ensuring continued access to their full potential. The SELLER reserves the right to withdraw the PoC solutions if the BUYER does not proceed with the purchase following the demonstration.

8.1 AUTOML PLATFORM

To drive Digital Transformation (DX) in the existing setup, Seller is pleased to propose the deployment of an "AutoML Platform" on the existing PDCMS server for a limited period as a Proof of Concept (PoC).

The objective of this engagement is to demonstrate the platform's ability to accelerate data-driven insights, streamline model development, and enhance decision-making capabilities without extensive manual intervention.

- The AutoML platform will be installed and configured on the existing PDCMS server infrastructure.
- The PoC will run for a limited duration solely for demonstration purposes.

The platform will enable: Automated data preprocessing, feature engineering, and model selection.

Model evaluation, and performance benchmarking.

Hyperparameter Tuning

Demonstration of ML Model working through predictive and prescriptive insights

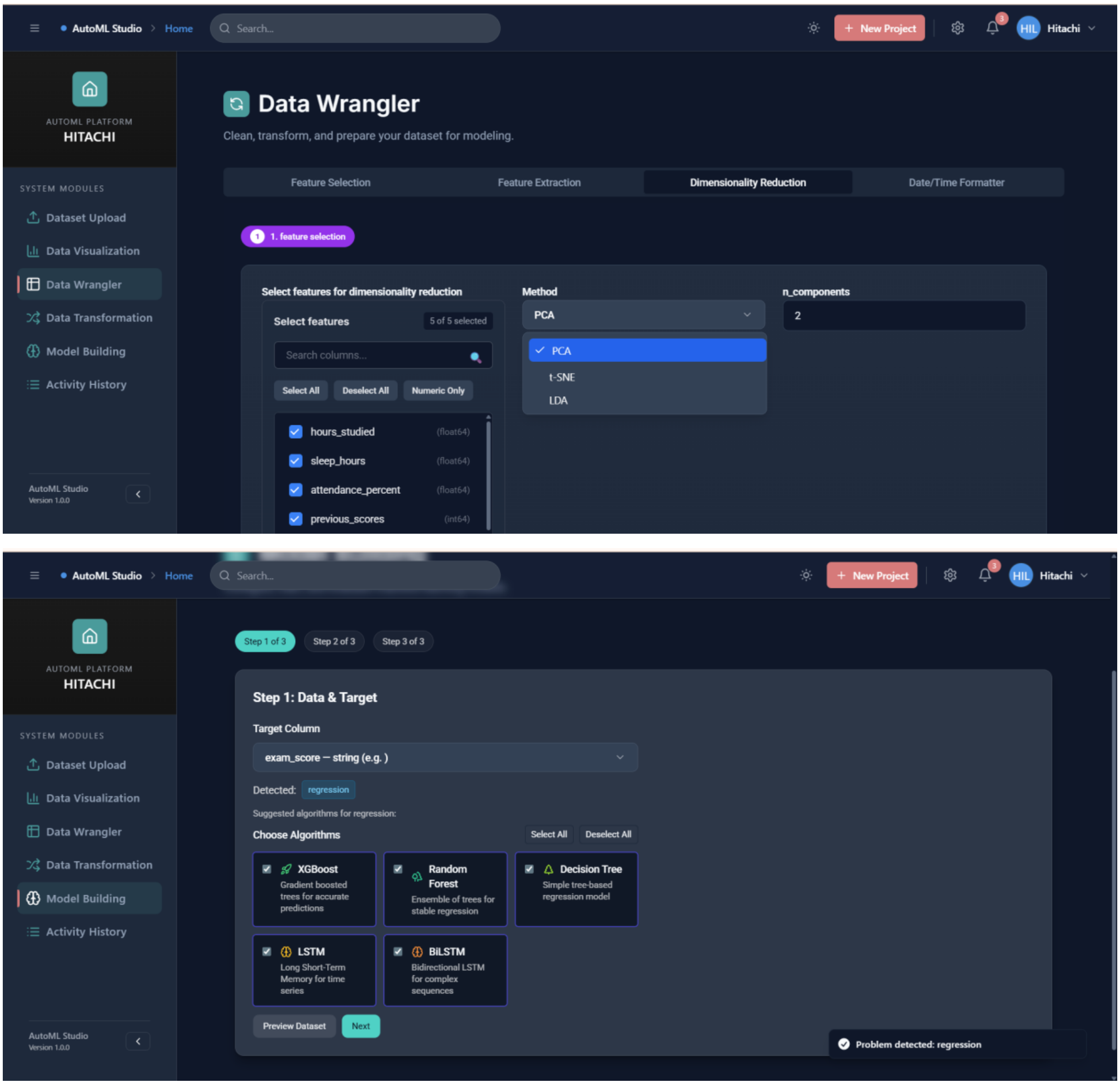


Fig: AutoML Platform Reference Screenshot

Note : *Seller shall not be bound with Performance guaranty & Remote Support for PoC Solutions.*

8.2 Email Reporting

This feature will send Shipping reports, log data, etc. to specified persons at a predefined frequency and timing. Detailed contents on this reporting feature will be finalized during engineering discussion and mutually agreed.

Automating reporting system can deliver substantial benefits to customer. Reporting system automation will reduce the time spent reporting and allow your team to do more process optimization and revenue generating activities. Reporting and email alert system ensures that you can deliver more comprehensive and customized reports to key executives, more quickly.

“Essential network configurations, tasks involved in connecting to the network, monitoring the network connection (including connection changes), giving users control over an app's network usage and authentications are to be decided and made/managed by buyer.”

8.3 SMS Alerts

It is important to take actions/decisions quickly to have desired effect. For this purpose, SMS alert system provides real-time updates to plant executive about significant events in the Yard. Contents of SMS and which events should trigger SMS alerts can be discussed and mutually agreed during detailed engineering discussion. SIM card will be Buyer Scope.

Note: Hardware required for carrying out POC will be discussed during Engineering stage.

End of Technical Technical Note